

PERSONAL INFORMATION

Hai Jiang

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🆔 ORCID [0009-0007-9735-7581](https://orcid.org/0009-0007-9735-7581)

Gender Male | Date of birth 21 December 1997 | Nationality Chinese

JOB APPLIED FOR Ph.D. Applicant

WORK EXPERIENCE

April 2026–Present Image Algorithm Engineer

Shenzhen Image Technology Co., Ltd.

- Develop C++ geometry algorithms for PCB copper-layer Gerber parsing and polygon feature extraction, converting raw contour data into structured inspection-ready representations.
- Extract edges, traces, corners, blocks, pads, and other manufacturing-critical geometric entities from large-scale polygon sets for downstream process inspection and quality analysis.
- Design computational geometry and topology analysis modules, optimize memory usage and throughput, and maintain unit tests and algorithm documentation for production use.

Skills C++, computational geometry, topology analysis, Gerber parsing, unit testing, technical documentation.

August 2025–April 2026 Computer Vision Algorithm Engineer

Shenzhen Zhongrui Power Technology Co., Ltd.

- Built an automated AI engine for a property-management data platform using YOLO, enabling object recognition, segmentation, and classification from user-uploaded images.
- Integrated OCR for receipts, signs, and document-like scenes, transforming detected text into structured outputs for platform workflows.
- Packaged the inference pipeline behind standardized APIs for web integration, using ONNX export and TensorRT acceleration to improve real-time response latency.

Skills YOLO, OCR, ONNX, TensorRT, API integration, computer vision deployment.

July 2022–July 2025 Research Assistant

Computational Medical Imaging Laboratory, Sun Yat-sen University

- Designed and validated algorithms for national medical AI projects covering MRI, dual-energy CT, and digital breast tomosynthesis.
- Built and evaluated deep learning models for clinical classification and detection tasks, supporting peer-reviewed manuscripts and translational research outcomes.
- Contributed to grant proposals, final reports, and industry collaboration across multidisciplinary medical-imaging projects.

Skills Medical imaging, deep learning, PyTorch, TensorFlow, data preprocessing, research writing.

EDUCATION

September 2019–July 2022 Master of Science in Computational Mathematics

M. Sc.

Nankai University (NKU, Project 985 & 211, Double First-Class), Tianjin, China

Thesis GANs-Based Personal Style Imitation of Chinese Handwritten Characters.

	Developed an end-to-end CycleGAN framework achieving 85% visual similarity (10% improvement over baselines).
Key Skills	GANs, style-transfer learning, Python, PyTorch, data pre-processing.
Supervisors	Prof. Yunhua Xue, Prof. Chunlin Wu.
Relevant Coursework	Approximation Theory, Numerical Optimization, Functional Analysis, Matrix Computation, Numerical PDEs.
GPA	77.4/100.0.

September 2014–July 2018 **Bachelor of Engineering in Information Security** B. Eng.

	Lanzhou University (LZU, Project 985 & 211, Double First-Class), Lanzhou, Gansu, China
Thesis	Improved Upper Bounds of Roman Domination Number in Maximal Outerplanar Graphs. Focused on graph theory and combinatorial optimization.
Supervisor	Prof. Zepeng Li.
Relevant Coursework	Discrete Mathematics, Data Structures, Operating Systems, C/C++ Programming, Database Theory.
GPA	4.15/5.00.

RESEARCH EXPERIENCE

July 2022–July 2025 **Computational Medical Imaging Laboratory** R. A.

School of Computer Science and Engineering, Sun Yat-sen University

Conducted three medical-imaging research projects spanning MRI-based placenta accreta spectrum classification, DECT-based metastasis prediction, and DBT multi-view detection.

- Project 1** Placenta Accreta Spectrum Disorder Classification
- Developed a prior-knowledge-based deep learning approach for PASD binary classification, achieving AUC of 0.9902.
 - Built a multi-task learning model using T2-WI MRI images for subtype classification, achieving AUC of 0.80.
 - Published: “Anatomy-Guided Multitask Learning for MRI-Based Classification of Placenta Accreta Spectrum and Its Subtypes” [1].
- Skills** Literature review, data preprocessing, PyTorch, research writing.
- Project 2** Breast Cancer Metastasis Prediction
- Designed a CNN-based framework using dual-energy CT scans to predict metastasis.
 - Achieved AUC of 0.85 (mean cross-validation AUC).
 - Manuscript submitted to MICCAI 2024 and under revision for Journal of Medical Physics [4].
- Skills** TensorFlow, Keras, data analysis, experimental design, research writing.
- Project 3** Multi-view Learning for Atypical Architectural Disorder Detection in Breast DBT
- Contributed to the multi-view learning framework for atypical architectural disorder detection in digital breast tomosynthesis (DBT) images.
 - Introduced a novel Siamese module with triplet loss to enhance feature extraction and detection performance [2].
- Skills** Mask-RCNN, Detection, Siamese Network, Multi-view, Triplet-Loss.

January 2022–June 2022 **Image Analysis Team**

School of Mathematical Sciences, Nankai University

- Project** ADMM Model for Compressive-Sensing MRI.
- Reproduced iterative mathematical equations from Deep ADMM-Net for Compressive-Sensing MRI using C++, Python, and PyTorch.
- Skills** Compressive-sensing theory, neural networks, MRI reconstruction.

PUBLICATIONS

- [1] **Hai Jiang**, Qiongtong Liu, Yuanpin Zhou, Jiawei Pan, Ting Song, and Yao Lu. "Anatomy-Guided Multitask Learning for MRI-Based Classification of Placenta Accreta Spectrum and its Subtypes". In: *2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*. 2025, pp. 1–5.
- [2] Jiawei Pan, Zilong He, Yue Li, Weixiong Zeng, Yaya Guo, Lixuan Jia, **Hai Jiang**, Weiguo Chen, and Yao Lu. "Atypical architectural distortion detection in digital breast tomosynthesis: a multi-view computer-aided detection model with ipsilateral learning". In: *Physics in Medicine & Biology* 68.23 (2023), p. 235006.
- [3] XueFang Wang, **Hai Jiang**, Ruxu Du, Yong Zhong, and Yao Lu. "Anatomical-Prior-Based Multiscale Segmentation of Cardiac Substructures Using Enhanced Skip-Connections and a Triple-View Fusion Network." In: *Submitting ()*.
- [4] **Hai Jiang**, Chushan Zheng, Jiawei Pan, Yuanpin Zhou, Qiongtong Liu, Xiang Zhang, Jun Shen, and Yao Lu. *DECT-based Space-Squeeze Method for Multi-Class Classification of Metastatic Lymph Nodes in Breast Cancer*. 2025. arXiv: 2505.17528 [eess.IV]. URL: <https://arxiv.org/abs/2505.17528>.
- [5] Jiawei Pan, Zilong He, Yue Li, Weixiong Zeng, Yaya Guo, Lixuan Jia, **Hai Jiang**, Weiguo Chen, and Yao Lu. "Multi-view Architectural Distortion Detection with Confidence Boosting in Digital Breast Tomosynthesis". In: *Submitted to MICCAI 2025 (rejected after rebuttal, currently under revision)*.

TECHNICAL SKILLS

Programming	Python, C/C++, PyTorch, TensorFlow, Keras, MATLAB, LaTeX, Git.
Tools & Platforms	Linux (Ubuntu), ONNX, TensorRT, CMake, Microsoft Office, Adobe Photoshop.
Research Methods	Deep learning, computer vision, image processing, medical imaging, computational geometry, compressive sensing.

AWARDS & SCHOLARSHIP

2014–2018	Four-time recipient of the Third-Class Merit Scholarship for Academic Excellence at LZU.
2019–2022	Three-time recipient of the Third-Class Merit Scholarship for Academic Excellence at NKU.

LANGUAGE PROFICIENCY

Mandarin	Native.
English	Professional (IELTS 7.0, CET6 476/710, CET4 544/710).
Cantonese	Intermediate.

RESEARCH PROJECT DOCUMENTATION

Funding Proposal Writing	Contributed to National Key R&D Program of China [No. 2023YFE0204300].
Report Writing	Contributed to completion and progress reports for National Natural Science Foundation of China [No. 81971691, 12126610] and R&D Program of Pazhou Lab [No. 2023K0606].

TEACHING EXPERIENCE

Courses Taught	Calculus, Mathematical Analysis.
Thesis Supervision	Breast Cancer Classification Method Based on Dual-Energy CT Images.

RESEARCH INTERESTS

Artificial Intelligence, Deep Learning (CNNs, GNNs).
Mathematics, Medical Image Analysis, Physics.

REFERENCES

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